

The European Space Agency (ESA; pronounced /'i:sə/ EE-sə) is a 23-member international organisation devoted to space exploration. It has its headquarters in Paris and a staff of around 3,000 people globally as of 2025. ESA was founded in 1975 in the context of European integration. Its 2026 annual budget was around €8.3 billion.

The ESA human spaceflight programme includes participation in the International Space Station (ISS) and collaboration with NASA on the Artemis programme, especially manufacturing of the Orion spacecraft's European Service Module (ESM). ESA launches and operates uncrewed missions to the Moon, Mars, Jupiter, Venus, Mercury, the Sun, and various comets and asteroids. Other activities include space telescopes, Earth observation satellites, asteroid impact avoidance, telecommunication and navigation satellites, designing launch vehicles (e.g. Ariane 6 is operated through Arianespace with ESA sharing in the costs), and maintaining Europe's spaceport (the Guiana Space Centre in Kourou, French Guiana), as well as space safety and commercialisation.

Mission

The treaty establishing the European Space Agency reads:

"The purpose of the Agency shall be to provide for and to promote, for exclusively peaceful purposes, cooperation among European States in space research and technology and their space applications, with a view to their being used for scientific purposes and for operational space applications systems."

ESA is responsible for setting a unified space and related industrial policy, recommending space objectives to the member states, and integrating national programmes like satellite development, into the European programme as much as possible.

Jean-Jacques Dordain—ESA's Director General (2003–2015)—outlined the European Space Agency's mission in a 2003 interview:

"Today space activities have pursued the benefit of citizens, and citizens are asking for a better quality of life on Earth. They want greater security and economic wealth, but they also want to pursue their dreams, to increase their knowledge, and they want younger people to be attracted to the pursuit of science and technology.

I think that space can do all of this: it can produce a higher quality of life, better security, more economic wealth, and also fulfill our citizens' dreams and thirst for knowledge, and attract the young generation. This is the reason space exploration is an integral part of overall space activities. It has always been so, and it will be even more important in the future."At the ministerial council of 2025, ESA member states widened the agency's mandate to include defence. The resolution reads:

"ESA's intergovernmental framework provides the credentials and tools for developing space technologies and systems [...] for security and defence and to implement corresponding activities."

History

After World War II, many European scientists left Western Europe to work with the United States. Although the 1950s boom made it possible for Western European countries to invest in research and specifically in space-related activities, Western European scientists realised solely national projects would not be able to compete with the two superpowers. In 1958, only months after the Sputnik shock, Edoardo Amaldi (Italy) and Pierre Auger (France), two prominent members of the Western European scientific community, met to discuss the foundation of a common Western European space agency. The meeting was attended by scientific representatives from eight countries.

The Western European nations decided to have two agencies: one concerned with developing a launch system, ELDO (European Launcher Development Organisation), of which Renzo

Carrobio di Carrobio became the first Secretary General from 1964 to 1971, and the other the precursor of the European Space Agency, ESRO (European Space Research Organisation), led by Pierre Auger. The latter was established on 20 March 1964 by an agreement signed on 14 June 1962. From 1968 to 1972, ESRO launched seven research satellites, but ELDO was not able to deliver a launch vehicle. Both agencies struggled with the underfunding and diverging interests of their participants.

Foundation

The ESA in its current form was founded with the ESA Convention in 1975, when ESRO was merged with ELDO. ESA had ten founding member states: Belgium, Denmark, France, West Germany, Italy, the Netherlands, Spain, Sweden, Switzerland, and the United Kingdom. These signed the ESA Convention in 1975 and deposited the instruments of ratification by 1980, when the convention came into force. During this interval the agency functioned in a de facto fashion.

First science missions

ESA launched its first major scientific mission in 1975, Cos-B, a satellite monitoring gamma-ray emissions in the universe, which was first worked on by ESRO. ESA collaborated with NASA on the International Ultraviolet Explorer (IUE), the world's first high-orbit telescope, which was launched in 1978 and operated successfully for 18 years. A number of successful Earth-orbit projects followed, and in 1986 ESA began Giotto, its first deep-space mission, to study the comets Halley and Grigg-Skjellerup. Hipparcos, a star-mapping mission, was launched in 1989 and in the 1990s SOHO, Ulysses, and the Hubble Space Telescope were all jointly carried out with NASA. Later scientific missions in cooperation with NASA include the Cassini-Huygens space probe, to which the ESA contributed by building the Titan landing module Huygens.

First launch vehicles

As the successor of ELDO, the ESA has also constructed rockets for scientific and commercial payloads. Ariane 1, launched in 1979, carried mostly commercial payloads into orbit from 1984 onward. The next two versions of the Ariane rocket family were intermediate stages in the development of a more advanced launch system, the Ariane 4, which operated between 1988 and 2003 and established the ESA as the world leader in commercial space launches in the 1990s. Although the succeeding Ariane 5 experienced a failure on its first flight in 1996, it has since firmly established itself within the heavily competitive commercial space launch market with 112 successful launches until 2023. In 1998, ESA started developing the small-lift launch vehicle Vega.

First astronauts

Astronauts from the future ESA member states have been joining Soviet space missions since 1978. The first ESA astronaut to fly to space was Ulf Merbold who joined the STS-9 mission of the American Space Shuttle in 1983, that carried the first European-built Spacelab laboratory module. The experience gained during the Spacelab programme was later instrumental in developing the International Space Station (ISS). Since then, ESA astronauts have been joining Space Shuttle flights, as well as Russian Soyuz flights to the Mir space station. During the 1980s and 1990s, ESA was considering developing its own crewed spacecraft Hermes and a small space station Columbus MTF. These plans were later abandoned and instead, ESA joined the ISS programme with Columbus being repurposed as one of the station's laboratory modules and ESA astronauts flying to the station on American and Russian spacecraft.

ESA in the 2000s

The beginning of the new millennium saw the ESA become, along with agencies like NASA and JAXA, one of the major participants in space research. Although ESA had relied on co-operation

with NASA in previous decades, changed circumstances, such as tough legal restrictions on information sharing under ITAR, led to decisions to rely more on itself and on cooperation with Roscosmos.

The agency continued its contribution to the International Space Station (ISS) programme with European astronauts joining assembly flights as well as long-term missions to the station. In 2008, ESA added its laboratory module Columbus to ISS and started launching the ATV cargo spacecraft. During the 2000s, ESA was considering cooperation with Russia on the proposed Kliper and CSTS crewed vehicles, but neither of these was developed.

Notable deep space missions during the 2000s included the agency's first Moon, Mars, and Venus orbiters: SMART-1, Mars Express, and Venus Express. ESA's Huygens probe, launched together with the NASA's Cassini mission in 1997, reached its destination in 2005 when it successfully landed on Titan, marking the farthest landing from Earth a spacecraft has ever made. The comet orbiter Rosetta launched in 2004 and performed multiple deep space flybys and observations during the decade, but wouldn't reach its destination until 2014.

ESA has launched multiple major astronomy missions in the 2000s: the gamma ray observatory INTEGRAL, the infrared observatory Herschel, the cosmic microwave background mapper Planck, and Corot, a milestone in the search for exoplanets. Notable Earth observation missions launched during the decade included Envisat, Double Star, GOCE, SMOS, and the experimental PROBA series. ESA also contributed to the meteorological constellations Meteosat and MetOp and tested technologies for the future Galileo satellite navigation system with two GIOVE satellites.

During the 2000s, ESA pursued the Aurora programme which planned to launch a series of increasingly ambitious missions culminating in a crewed landing on Mars. The programme was eventually abandoned with only the ExoMars series of astrobiology space probes to Mars remaining active. ExoMars, as well as the plans for human spaceflight to LEO, the Moon, and Mars, were later incorporated into the new Terrae Novae programme established in 2016.

ESA in the 2010s

In 2010, ESA added the Cupola observation module to ISS and European astronauts continued joining long-term missions to the station. The ATV cargo spacecraft continued resupplying ISS until its last flight in 2015. In 2012, ESA committed to providing the ATV-derived European Service Module for NASA's crewed lunar spacecraft Orion.

In 2014, ESA's Rosetta probe arrived at its destination, the Jupiter-family comet 67P/Churyumov-Gerasimenko. It became the first spacecraft ever to orbit a comet and its lander Philae performed the first ever landing on a comet. In 2016, ESA launched its second Mars orbiter mission, the ExoMars Trace Gas Orbiter (TGO), as the first ExoMars mission within the newly established Terra Nova programme. When the spacecraft arrived at Mars later the same year, it released the Schiaparelli lander, which failed on landing. TGO, however, entered the Martian orbit and after 11 months of aerobraking began its scientific observations, focused mostly on the atmosphere of Mars. In 2018, ESA and JAXA launched the joint mission BepiColombo, which is expected to arrive at Mercury in 2026.

Notable astronomy missions launched in 2010s were the astrometry telescope Gaia, which produced the largest and most precise 3D catalogue of astronomical objects ever made, and the exoplanets-characterising telescope CHEOPS. ESA also launched LISA Pathfinder, a technology demonstrator for the future gravitational wave observatory LISA. In 2010, ESA launched the cryosphere-monitoring satellite CryoSat-2, a replacement for CryoSat-1 which had been destroyed in 2005 due to a failure of its Russian launch vehicle. Another major Earth observation satellite mission of the decade was the magnetic field-observing Swarm, launched in 2013.

The 2010s saw the first launches of two major European satellite constellations, to which ESA contributed alongside other European institutions, the satellite navigation system Galileo and the Earth observation programme Copernicus with its Sentinel satellites. The first operational

pair of Galileo satellites was launched in 2011. The radar satellite Sentinel-1A, first dedicated mission of the Copernicus Programme, was launched in 2014, followed by the optical imaging satellite Sentinel-2A in 2015, the oceanography satellite Sentinel-3A in 2016, and the air pollution-monitoring Sentinel-5p in 2017.

The small-lift launch vehicle Vega, developed by the Italian company Avio, had its first flight in 2012 and then flew 22 times until 2024 when it was replaced by the more powerful Vega-C. In 2016, ESA started supporting the Spanish company PLD Space with FLPP funding for the development of their commercial reusable launch vehicle Miura 5.

ESA in the 2020s

After the 2022 Russian invasion of Ukraine, the cooperation between ESA and Roscosmos was mostly severed. This led to a delay in the ExoMars programme and ending of Soyuz launches from the Guiana Space Centre. Furthermore, the space policy of the second Trump administration brought uncertainty into many joint ESA-NASA programmes including major science missions and human spaceflight. The ministerial council held in November 2025 reacted to the changed international situation by including security and defence in ESA's mandate, funding development of new European launch vehicles, as well as confirming many priorities of the long term, autonomy-focused "Strategy 2040" proposed earlier in 2025 and approving a record budget of €22.1 billion. In early 2026, ESA established the new Resilience, Navigation, and Connectivity Directorate focused on security and defence technologies.

In early 2020s, ESA has added two major components to the ISS. The Bartolomeo platform was connected to the Columbus module in 2020 to increase its capacity for external payloads. The European Robotic Arm was launched in 2021 together with the Nauka module. ESA continued contributing European astronauts to regular ISS expeditions, and also started paying private companies for short-term astronaut flights to the station. ESA also started supporting European companies in developing uncrewed space capsules for resupplying the ISS and

future space stations via the LCRS initiative, as well as the Indian ISRO in developing their crewed Gaganyaan spacecraft. The ESA-built European Service Module of the Orion spacecraft flew on its first uncrewed test flight in 2022 and in 2026, it brought Artemis II, the first beyond-low Earth orbit crewed spaceflight since 1972, around the Moon.

The BepiColombo mission, launched in 2018, has completed all its nine gravity assist manoeuvres at Earth, Venus, and Mercury, in preparation for the planned 2026 insertion into Mercury orbit. The heliophysics mission Solar Orbiter, launched in 2020, continues to perform periodic Venus flybys which gradually increase its orbital inclination, allowing it to observe the Sun from outside the Solar system plane. The Juice mission, launched in 2023, has so far completed two gravity assist manoeuvres at Earth and Venus on its way to enter an orbit around Jupiter in 2031. Two European orbiters, Mars Express and TGO, continued their operations around Mars, providing scientific insights into the planet's surface and atmosphere. In 2025, Mars Express received a software update, which could allow it to stay operational until 2030s.

The Hera mission, launched in 2024 as the first space mission of the optional Space Safety Programme, flew by Mars in 2025 on its way to perform a post-impact survey of the asteroid Dimorphos which had been impacted by NASA's Double Asteroid Redirection Test mission. The Space Safety Programme was established in 2019 by substantially expanding the former Space Situational Awareness (SSA) programme. Since then, it prepared an extensive series of space missions and ground-based projects focused on mitigating the dangers of near-Earth asteroids, space weather, and space debris, which was fully endorsed by ESA member states.

The long-awaited NASA-ESA-CSA space telescope, the James Webb Space Telescope, finally launched in 2021 and started operating in its halo orbit around the Sun–Earth L2 point in 2022. The first half of 2023 saw the launch the Euclid space telescope designed to better understand dark energy and dark matter by accurately measuring the accelerating expansion of the universe. The experimental PROBA-3 mission, launched in 2024, successfully

demonstrated high-precision formation flying for coronagraphy observation of the Sun. The space weather mission SMILE, developed in cooperation with CAS, was launched in May 2026. The successor to the Vega launch vehicle, Vega C, launched successfully for the first time in July 2022, delivering the LARES 2 satellite and six CubeSats to Earth orbit. The successor to the Ariane 5 launch vehicle, Ariane 6, had its maiden flight in July 2024, followed by the first commercial launch in March 2025 and the first launch of the 4-booster variant Ariane 64 in February 2026. In the 2020s, ESA started supporting aspiring European commercial launch providers with substantial development funding and launch contracts through the Boost! programme, the Flight Ticket Initiative, and the European Launcher Challenge (ELC).

Facilities

The agency's facilities date back to ESRO and are deliberately distributed among various countries and areas. The most important are the following centres:

ESA headquarters in Paris, France;

ESA science missions are based at ESTEC in Noordwijk, Netherlands;

Earth Observation missions at the ESA Centre for Earth Observation in Frascati, Italy;

ESA Mission Control (ESOC) is in Darmstadt, Germany;

The European Astronaut Centre (EAC) that trains astronauts for future missions is situated in Cologne, Germany;

The European Centre for Space Applications and Telecommunications (ECSAT), a research institute created in 2009, is located in Harwell, England, United Kingdom;

The European Space Astronomy Centre (ESAC) is located in Villanueva de la Cañada, Madrid, Spain.

The European Space Security and Education Centre (ESEC), located in Redu, Belgium;

The ESTRACK tracking and deep space communication network.

Many other facilities are operated by national space agencies in close collaboration with ESA.

Esrange near Kiruna in Sweden;

Guiana Space Centre in Kourou, France;

Toulouse Space Centre, France;

Institute of Space Propulsion in Lampoldshausen, Germany;

Columbus Control Centre in Oberpfaffenhofen, Germany.

Programmes

Mandatory programmes

Every member state must contribute to the mandatory programmes. The contributions to the mandatory programmes are calculated based on the states' Gross National Product. Decisions about these programmes are taken unanimously.

European Space Agency Science Programme

Horizon 2000

Horizon 2000 Plus

Cosmic Vision

Voyage 2050

Technology Development Element Programme

Science Core Technology Programme

Optional programmes

Depending on their individual choices, the member states can contribute to optional programmes.

Launchers

Ariane rocket family

Vega, Vega-C

Future Launchers Preparatory Programme

Earth Observation

FutureEO (formerly Living Planet Programme)

Copernicus

MetOp-SG

Human Spaceflight and Exploration

Aurora

Terrae Novae (formerly European Exploration Envelope Programme, E3P)

Telecommunications (managed by ESA Connectivity and Secure Communications directorate)

Advanced Research in Telecommunication Systems (ARTES)

IRIS2

HydRON

Moonlight Programme

Navigation

EGNOS

Galileo

Space Safety Programme (formerly Space Situational Awareness)

Technology

General Support Technology Programme (GSTP)

Member states, funding, and budget

Membership and contribution to the ESA

Member states participate to varying degrees with both mandatory space programmes and those that are optional. As of 2008, the mandatory programmes made up 25% of total expenditures while optional space programmes were the other 75%. The ESA has traditionally

implemented a policy of "georeturn", where funds that ESA member states provide to the ESA "are returned in the form of contracts to companies in those countries."

By 2015, the ESA was an intergovernmental organisation of 22 member states. The 2008 ESA budget amounted to €3.0 billion whilst the 2009 budget amounted to €3.6 billion. The total budget amounted to about €3.7 billion in 2010, €3.99 billion in 2011, €4.02 billion in 2012, €4.28 billion in 2013, €4.10 billion in 2014, €4.43 billion in 2015, €5.25 billion in 2016, €5.75 billion in 2017, €5.60 billion in 2018, €5.72 billion in 2019, €6.68 billion in 2020, €6.49 billion in 2021, €7.15 billion in 2022, €7.46 billion in 2023 and €7.79 billion in 2024.

English and French are the two official languages of the ESA. Additionally, official documents are also provided in German and documents regarding the Spacelab have been also provided in Italian. If found appropriate, the agency may conduct its correspondence in any language of a member state.

The following table lists all the member states and adjunct members, their ESA convention ratification dates, and their contributions as of 2026:

Non-full member states

Previously associated members were Austria, Norway, Finland and Slovenia, all of which later joined the ESA as full members. Since March 2026 there have been five associate members: Latvia, Lithuania, Slovakia, Cyprus and Canada. The four European members have shown interest in full membership and may eventually apply within the next years.

Latvia

Latvia became the second current associated member on 30 June 2020, when the Association Agreement was signed by ESA Director Jan Wörner and the Minister of Education and Science of Latvia, Ilga Šuplinska in Riga. The Saeima ratified it on 27 July.

Lithuania

In May 2021, Lithuania became the third current associated member. As a consequence its citizens became eligible to apply to the 2022 ESA Astronaut group, applications for which were scheduled to close one week later. The deadline was therefore extended by three weeks to allow Lithuanians a fair chance to apply.

Slovakia

Slovakia's Associate membership came into effect on 13 October 2022, for an initial duration of seven years. The Association Agreement supersedes the European Cooperating State (ECS) Agreement, which entered into force upon Slovakia's subscription to the Plan for European Cooperating States Charter on 4 February 2016, a scheme introduced at ESA in 2001. The ECS Agreement was subsequently extended until 3 August 2022.

Cyprus

Cyprus has signed an Associate Agreement in October 2025, with expectations of becoming an associate member by early 2026. On 19 March 2026 Cyprus officially became an Associate member of ESA.

Canada

Since 1 January 1979, Canada has had the special status of a Cooperating State within the ESA. By virtue of this accord, the Canadian Space Agency takes part in the ESA's deliberative bodies and decision-making and also in the ESA's programmes and activities. Canadian firms can bid for and receive contracts to work on programmes. The accord has a provision ensuring a fair industrial return to Canada. The most recent Cooperation Agreement was signed on 12 February 2019 with a term extending to 2030.

For 2014, Canada's annual assessed contribution to the ESA general budget was €6,059,449

(CAD\$8,559,050). This annual contribution increased in 2017 to €21,600,000 (CAD\$30,000,000). In 2025, the Government of Canada announced a plan to increase contributions to the ESA by €326,000,000 (CAD\$528,500,000) over the following three to five-year period.

On 14 April 2026 ESA and Canada signed a General Security of Information Agreement (GSOIA), which establishes a legal framework for the exchange of classified information. This was intended to facilitate collaboration in strategic domains such as surveillance, disaster response, and security and to support the development of dual-use technologies.

Budget appropriation and allocation

The ESA is funded from annual contributions by national governments of members as well as from an annual contribution by the European Union (EU).

The budget of the ESA was €5.250 billion in 2016. Every 3–4 years, ESA member states agree on a budget plan for several years at an ESA member states conference. This plan can be amended in future years, however provides the major guideline for the ESA for several years. The 2016 budget allocations for major areas of the ESA activity are shown in the chart on the right.

Countries typically have their own space programmes that differ in how they operate organisationally and financially with the ESA. For example, the French space agency CNES has a total budget of €2,015 million, of which €755 million is paid as direct financial contribution to the ESA. Several space-related projects are joint projects between national space agencies and the ESA (e.g. COROT). Also, the ESA is not the only European governmental space organisation (for example European Union Satellite Centre and the European Union Space Programme Agency).

Enlargement

After the decision of the ESA Council of 21/22 March 2001, the procedure for accession of the

European states was detailed as described the document titled "The Plan for European Co-operating States (PECS)". Nations that want to become a full member of the ESA do so in 3 stages. First a Cooperation Agreement is signed between the country and ESA. In this stage, the country has very limited financial responsibilities. If a country wants to co-operate more fully with ESA, it signs a European Cooperating State (ECS) Agreement, albeit to be a candidate for said agreement, a country must be European. The ECS Agreement makes companies based in the country eligible for participation in ESA procurements. The country can also participate in all ESA programmes, except for the Basic Technology Research Programme. While the financial contribution of the country concerned increases, it is still much lower than that of a full member state. The agreement is normally followed by a Plan For European Cooperating State (or PECS Charter). This is a 5-year programme of basic research and development activities aimed at improving the nation's space industry capacity. At the end of the 5-year period, the country can either begin negotiations to become a full member state or an associated state or sign a new PECS Charter. Many countries, most of which joined the EU in both 2004 and 2007, have started to co-operate with the ESA on various levels:

During the Ministerial Meeting in December 2014, ESA ministers approved a resolution calling for discussions to begin with Israel, Australia and South Africa on future association agreements. The ministers noted that "concrete cooperation is at an advanced stage" with these nations and that "prospects for mutual benefits are existing".

A separate space exploration strategy resolution calls for further co-operation with the United States, Russia and China on "LEO exploration, including a continuation of ISS cooperation and the development of a robust plan for the coordinated use of space transportation vehicles and systems for exploration purposes, participation in robotic missions for the exploration of the Moon, the robotic exploration of Mars, leading to a broad Mars Sample Return mission in which Europe should be involved as a full partner, and human missions beyond LEO in the

longer term."

In August 2019, the ESA and the Australian Space Agency signed a joint statement of intent "to explore deeper cooperation and identify projects in a range of areas including deep space, communications, navigation, remote asset management, data analytics and mission support". Details of the cooperation were laid out in a framework agreement signed by the two entities. On 17 November 2020, ESA signed a memorandum of understanding (MOU) with the South African National Space Agency (SANSA). SANSA CEO Dr. Valanathan Munsami tweeted: "Today saw another landmark event for SANSA with the signing of an MoU with the ESA. This builds on initiatives that we have been discussing for a while already and which gives effect to these. Thanks Jan for your hand of friendship and making this possible."

Launch vehicles

The ESA currently has two operational launch vehicles Vega C and Ariane 6. Rocket launches are carried out by Arianespace, which has 23 shareholders representing the industry that manufactures the Ariane 5 as well as CNES, at the ESA's Guiana Space Centre. Because many communication satellites have equatorial orbits, launches from French Guiana are able to take larger payloads into space than from spaceports at higher latitudes. In addition, equatorial launches give spacecraft an extra 'push' of nearly 500 m/s due to the higher rotational velocity of the Earth at the equator compared to near the Earth's poles where rotational velocity approaches zero. In 2014, ESA proposed changes to the Ariane family development scheme by moving to competitive bids for the development of Ariane 6.

Ariane 6

Ariane 6 is a heavy lift expendable launch vehicle developed by Arianespace. The Ariane 6 entered into its inaugural flight campaign on 26 April 2024 with the flight conducted on 9 July 2024.

Vega-C

Vega is the ESA's carrier for small satellites. Developed by seven ESA members led by Italy. It is capable of carrying a payload with a mass of between 300 and 1500 kg to an altitude of 700 km, for low polar orbit. Its maiden launch from Kourou was on 13 February 2012. Vega began full commercial exploitation in December 2015.

The rocket has three solid propulsion stages and a liquid propulsion upper stage (the AVUM) for accurate orbital insertion and the ability to place multiple payloads into different orbits.

A larger version of the Vega launcher, Vega-C had its first flight in July 2022. The new evolution of the rocket incorporates a larger first stage booster, the P120C replacing the P80, an upgraded Zefiro second stage, and the AVUM+ upper stage. This new variant enables larger single payloads, dual payloads, return missions, and orbital transfer capabilities.

Future rocket development

Future projects under development within the Future Launchers Preparatory Programme (FLPP) include the Prometheus reusable engine technology demonstrator, Phoebus (an upgraded second stage for Ariane 6), and Themis (a reusable first stage).

Human spaceflight

Formation and development

At the time the ESA was formed, its main goals did not encompass human space flight; rather it considered itself to be primarily a scientific research organisation for uncrewed space exploration in contrast to its American and Soviet counterparts. It is therefore not surprising that the first non-Soviet European in space was not an ESA astronaut on a European space craft; it was Czechoslovak Vladimír Remek who in 1978 became the first non-Soviet or American in space (the first man in space being Yuri Gagarin of the Soviet Union) – on a Soviet

Soyuz spacecraft, followed by the Pole Mirosław Hermaszewski and East German Sigmund Jähn in the same year. This Soviet co-operation programme, known as Intercosmos, primarily involved the participation of Eastern bloc countries. In 1982, however, Jean-Loup Chrétien became the first non-Communist Bloc astronaut on a flight to the Soviet Salyut 7 space station.

Because Chrétien did not officially fly into space as an ESA astronaut, but rather as a member of the French CNES astronaut corps, the German Ulf Merbold is considered the first ESA astronaut to fly into space. He participated in the STS-9 Space Shuttle mission that included the first use of the European-built Spacelab in 1983. STS-9 marked the beginning of an extensive ESA/NASA joint partnership that included dozens of space flights of ESA astronauts in the following years. Some of these missions with Spacelab were fully funded and organisationally and scientifically controlled by the ESA (such as two missions by Germany and one by Japan) with European astronauts as full crew members rather than guests on board. Beside paying for Spacelab flights and seats on the shuttles, the ESA continued its human space flight co-operation with the Soviet Union and later Russia, including numerous visits to Mir.

During the latter half of the 1980s, European human space flights changed from being the exception to routine and therefore, in 1990, the European Astronaut Centre in Cologne, Germany was established. It selects and trains prospective astronauts and is responsible for the co-ordination with international partners, especially with regard to the International Space Station. As of 2006, the ESA astronaut corps officially included twelve members, including nationals from most large European countries except the United Kingdom.

In 2008, the ESA started to recruit new astronauts so that final selection would be due in spring 2009. Almost 10,000 people registered as astronaut candidates before registration ended in June 2008. 8,413 fulfilled the initial application criteria. Of the applicants, 918 were chosen to take part in the first stage of psychological testing, which narrowed down the field

to 192. After two-stage psychological tests and medical evaluation in early 2009, as well as formal interviews, six new members of the European Astronaut Corps were selected – five men and one woman.

Crew vehicles

In the 1980s, France pressed for an independent European crew launch vehicle. Around 1978, it was decided to pursue a reusable spacecraft model and starting in November 1987 a project to create a mini-shuttle by the name of Hermes was introduced. The craft was comparable to early proposals for the Space Shuttle and consisted of a small reusable spaceship that would carry 3 to 5 astronauts and 3 to 4 metric tons of payload for scientific experiments. With a total maximum weight of 21 metric tons it would have been launched on the Ariane 5 rocket, which was being developed at that time. It was planned solely for use in low Earth orbit space flights. The planning and pre-development phase concluded in 1991; the production phase was never fully implemented because at that time the political landscape had changed significantly. With the fall of the Soviet Union, the ESA looked forward to co-operation with Russia to build a next-generation space vehicle. Thus the Hermes programme was cancelled in 1995 after about 3 billion dollars had been spent. The Columbus space station programme had a similar fate.

In the 21st century, ESA started new programmes to create its own crew vehicles, most notable among its various projects and proposals is Hopper, whose prototype by EADS, called Phoenix, has already been tested. While projects such as Hopper are neither concrete nor to be realised within the next decade, other possibilities for human spaceflight in co-operation with Roscosmos (the Russian space agency) have emerged. Following talks with Roscosmos in 2004 and June 2005, a co-operation between the ESA and Roscosmos was announced to jointly work on the Russian-designed Kliper, a reusable spacecraft that would be available for space travel beyond LEO (e.g. the moon or even Mars). It was speculated that Europe would finance

part of it. A€50 million participation study for Kliper, which was expected to be approved in December 2005, was finally not approved by ESA member states. The Russian state tender for the project was subsequently cancelled in 2006.

In June 2006, ESA member states granted 15 million to the Crew Space Transportation System (CSTS) study, a two-year study to design a spacecraft capable of going beyond Low-Earth orbit based on the current Soyuz design. This project was pursued with Roskosmos instead of the cancelled Kliper proposal. A decision on the actual implementation and construction of the CSTS spacecraft was contemplated for 2008. In mid-2009 EADS Astrium was awarded a €21 million study into designing a crew vehicle based on the European ATV which was believed to be the basis of the Advanced Crew Transportation System design. Neither of these projects was pursued further in the 2010s.

In November 2012, ESA decided to join NASA's Orion programme. The ATV would form the basis of the European Service Module (ESM) for NASA's new crewed spacecraft. ESA may also seek to work with NASA on Orion's launch system as well to secure a seat on the spacecraft for its own astronauts. The complete Orion with ESM flew on its first mission to Lunar orbit in 2022.

In September 2014, the ESA signed an agreement with Sierra Nevada Corporation for co-operation in Dream Chaser project. Further studies on the Dream Chaser for European Use or DC4EU project were funded, including the feasibility of launching a Europeanised Dream Chaser onboard Ariane 5.

International Space Station (ISS)

With regard to the International Space Station (ISS), the ESA is not represented by all of its member states: 11 of the 22 ESA member states currently participate in the project: Belgium, Denmark, France, Germany, Italy, Netherlands, Norway, Spain, Sweden, Switzerland, and the United Kingdom. Austria, Finland and Ireland chose not to participate, because of lack of

interest or concerns about the expense of the project. Portugal, Luxembourg, Greece, the Czech Republic, Romania, Poland, Estonia, and Hungary joined ESA after the agreement had been signed.

ESA takes part in the construction and operation of the ISS, with contributions such as Columbus, a science laboratory module that was brought into orbit by NASA's STS-122 Space Shuttle mission, and the Cupola observatory module that was completed in July 2005 by Alenia Spazio for the ESA. The current estimates for the ISS are approaching €100 billion in total (development, construction and 10 years of maintaining the station) of which the ESA has committed to paying €8 billion. About 90% of the costs of the ESA's ISS share will be contributed by Germany (41%), France (28%) and Italy (20%). German ESA astronaut Thomas Reiter was the first long-term ISS crew member.

ESA has developed the Automated Transfer Vehicle for ISS resupply. Each ATV has a cargo capacity of 7,667 kilograms (16,903 lb). The first ATV, Jules Verne, was launched on 9 March 2008 and on 3 April 2008 successfully docked with the ISS. This manoeuvre, considered a major technical feat, involved using automated systems to allow the ATV to track the ISS, moving at 27,000 km/h, and attach itself with an accuracy of 2 cm. Five vehicles were launched before the programme ended with the launch of the fifth ATV, Georges Lemaître, in 2014.

European Life and Physical Sciences research on board the International Space Station (ISS) is mainly based on the European Programme for Life and Physical Sciences in Space programme that was initiated in 2001.

CubeSats

Since the 2010s, ESA has launched numerous CubeSat-type small satellites supported through various programmes:

Plan for Cooperating States

ESTCube-1, launched in 2013

General Support Technology Programme (GSTP)

GSTP also supports other non-CubeSat small satellite missions, e.g. MicroHETSat or the PROBA series.

GOMX-3, launched in 2015

GOMX-4B, launched in 2018

QARMAN, launched in 2019

OPS-SAT, launched in 2019

SIMBA, launched in 2020

PICASSO, launched in 2020

RadCube, launched in 2021

Sunstorm, launched in 2021

PRETTY, launched in 2023

PROBA-V CubeSat Companion, launched in 2023

GENA-OT, launched in 2025

CubeSpec, launching in 2026

GOMX-5, launching in 2026

HENON, launching in 2027

e.Inspector, launching in 2027

VULCAIN, launching in 2027

LUMIO, launching in 2027

M-ARGO, launching in 2027

AOS-P and AOS-D, launching in 2028

SROC, launching with Space Rider in 2028

Fly Your Satellite! (FYS)

FYS is an educational programme for student teams developing CubeSats or PocketQubes

AAUSAT5, deployed from the ISS in 2015 (FYS Pilot Edition)

e-sta@r-II, OUFTI-1, and AAUSAT4, launched in 2016 (FYS First Edition)

LEDSAT, launched in 2021

EIRSAT-1, launched in 2023

ISTSat-1 and 3Cat-4, launched in 2024

FrontierSat, launched in 2026

CIMER, launching in 2027

FutureEO Programme

Phi-Sat-1, launched in 2020

Intuition-1, launched in 2023

Phi-Sat-2, launched in 2024

InCubed

The InCubed co-funding programme, managed by ESA's Φ -lab, supports innovative technologies for Earth observation satellites.

MANTIS, launched in 2023

Hyperfield, hyperspectral imaging constellation, first satellite launched in 2024

AIX, launched in 2025

HiVE, the High-precision Versatile Ecosphere constellation, first satellite launched in 2025

FOREST-3, launched in 2025

TALISMAN, launching in 2026

Space Safety Programme (S2P)

Juventas and Milani, launched in 2024 together with Hera

PRELUDE, launching in 2027

Farinella and Don Quijote, launching in 2028 together with Ramses

Satis, launch planned for 2030

Greek CubeSat In-Orbit Validation

DUTHSat-2, launched in 2025

MICE-1, launched in 2025

PHASMA (LAMARR and DIRAC), launched in 2025

ERMIS (3 CubeSats), launched in 2026

OptiSat, launched in 2026

PeakSat, launched in 2026

Hellenic Space Dawn (Helios and Selene), launched in 2026

Pioneer Partnership Projects

Pioneer is part of ESA's Advanced Research in Telecommunications Systems (ARTES) programme.

Spire Global SaaS, launched in 2026

VIREON (2 CubeSats), launched in 2026

In-Orbit Demonstration and Validation (IOD/IOV)

IOD/IOV is a European Union programme funded through Horizon 2020 and entrusted to ESA.

IOD/IOV also supports other non-CubeSat small satellite missions, e.g. SYNDEO-3.

Tom & Jerry

CASSINI IHE1-1

IHE-1-4

IHE1-5

Small Missions for Exploration

VMMO, launching in 2028

Cybersecurity Operations Centre (CSOC)

CyberCUBE, launched in 2026

Cooperation with other countries and organisations

The ESA has signed co-operation agreements with the following states that currently neither plan to integrate as tightly with ESA institutions as Canada, nor envision future membership of the ESA: Argentina, Brazil, China, India, Russia, and Turkey. Additionally, the ESA has joint projects with the EUSPA of the European Union, NASA of the United States and is participating in the International Space Station together with the United States (NASA), Russia (Roscosmos), Japan (JAXA), and Canada (CSA).

National space organisations of member states

The Centre National d'Études Spatiales (CNES) (National Centre for Space Study) is the French government space agency (administratively, a "public establishment of industrial and commercial character"). Its headquarters are in central Paris. CNES is the main participant on the Ariane project. Indeed, CNES designed and tested all Ariane family rockets (mainly from its

centre in Évry near Paris)

The UK Space Agency is a partnership of the UK government departments which are active in space. Through the UK Space Agency, the partners provide delegates to represent the UK on the various ESA governing bodies. Each partner funds its own programme.

The Italian Space Agency (Agenzia Spaziale Italiana or ASI) was founded in 1988 to promote, co-ordinate and conduct space activities in Italy. Operating under the Ministry of the Universities and of Scientific and Technological Research, the agency cooperates with numerous entities active in space technology and with the president of the Council of Ministers. Internationally, the ASI provides Italy's delegation to the Council of the European Space Agency and to its subordinate bodies.

The German Aerospace Center (DLR) (German: Deutsches Zentrum für Luft- und Raumfahrt e. V.) is the national research centre for aviation and space flight of the Federal Republic of Germany and of other member states in the Helmholtz Association. Its extensive research and development projects are included in national and international cooperative programmes. In addition to its research projects, the centre is the assigned space agency of Germany bestowing headquarters of German space flight activities and its associates.

The Instituto Nacional de Técnica Aeroespacial (INTA) (National Institute for Aerospace Technique) is a Public Research Organisation specialised in aerospace research and technology development in Spain. Among other functions, it serves as a platform for space research and acts as a significant testing facility for the aeronautic and space sector in the country.

NASA

The ESA has a long history of collaboration with NASA. Since ESA's astronaut corps was formed, the Space Shuttle has been the primary launch vehicle used by the ESA's astronauts to get into space through partnership programmes with NASA. In the 1980s and 1990s, the

Spacelab programme was an ESA-NASA joint research programme that had the ESA develop and manufacture orbital labs for the Space Shuttle for several flights in which the ESA participates with astronauts in experiments.

In October 2020, ESA entered into a memorandum of understanding (MOU) with NASA to collaborate on the Artemis program, which would provide an orbiting Lunar Gateway and also accomplish the first crewed lunar landing in 50 years. Gateway was later cancelled by NASA.

ESA purchases seats on the NASA operated Commercial Crew Program. The first ESA astronaut to be on a Commercial Crew Program mission was Thomas Pesquet who launched into space aboard Crew Dragon Endeavour on the Crew-2 mission. Following ESA astronauts on Crew Dragon missions were Matthias Maurer (Crew-3), Samantha Cristoforetti (Crew-4), and others.

In robotic science mission and exploration missions, NASA has been the ESA's main partner. Cassini-Huygens was a joint NASA-ESA mission, along with the Infrared Space Observatory, INTEGRAL, SOHO, and others. The Hubble Space Telescope and James Webb Space Telescope are joint projects of NASA and ESA. Possible future ESA-NASA joint projects include the Laser Interferometer Space Antenna, Mars sample-return mission, and others.

JAXA

ESA and the Japanese Space Agency (JAXA) have engaged in long-standing collaboration in various domains of space exploration, satellite technology, space science, and human spaceflight. Their partnership has evolved over the decades, focusing on joint missions, technology sharing, and international cooperation in space exploration. Some of the programmes include BepiColombo Mission, EarthCARE, Hera, and Comet Interceptor. In 2025, ESA opened its first Asian office in Tokyo, Japan.

Roscosmos

In the 2000s and 2010s, ESA collaborated with Roscosmos (the Russian space agency) on the development of the crewed vehicle CSTS (later cancelled), the use of French Guiana's Guiana Space Centre spaceport for launches of Soyuz-2 rockets, and the ExoMars programme including the Schiaparelli lander and the Trace Gas Orbiter. In 2022, most collaboration with the Roscosmos was terminated after the 2022 Russian invasion of Ukraine.

CNSA

ESA and the Chinese Space Agency (CNSA) cooperated on development of the Double Star Mission. In 2017, ESA sent two astronauts to China for two weeks sea survival training with Chinese astronauts in Yantai, Shandong. In 2024, the joint Einstein Probe was launched. The joint ESA-CAS mission SMILE was launched in 2026 to image for the first time the magnetosphere of the Sun in soft X-rays and UV.

ISRO

ESA provided instruments for the Indian Space Research Organisation (ISRO)'s lunar mission Chandrayaan-1 in 2008. Since 2024, ESA has been supporting ISRO's human spaceflight programme Gaganyaan.

AfSA

ESA has been cooperating with the African Union's African Space Agency (AfSA) since its formation (itself inspired by ESA) in 2023.

Link between ESA and EU

The ESA is an independent space agency and not under the jurisdiction of the European Union, although they have common goals, share funding, and work together often. The initial aim of the European Union (EU) was to make the European Space Agency an agency of the EU

by 2014. While the EU and its member states fund together 86% of the budget of the ESA, it is not an EU agency. Furthermore, the ESA has several non-EU members, most notably the United Kingdom which left the EU while remaining a full member of the ESA. The ESA is partnered with the EU on two space programmes, the Copernicus series of Earth observation satellites and the Galileo satellite navigation system. The ESA provided technical oversight for both as well as funding for Copernicus. The EU has shown interest in expanding into space exploration with a proposal to rename and further develop its satellite navigation agency (the European GNSS Agency) into the new EU Agency for the Space Programme. The ESA strongly criticized the proposal, as it was seen as competition or replacement of the ESA.

In January 2021, after years of acrimonious relations, EU and ESA officials mended their relationship, with the EU Internal Market commissioner Thierry Breton saying "The European space policy will continue to rely on the ESA and its unique technical, engineering and science expertise," and that the "ESA will continue to be the European agency for space matters. If we are to be successful in our European strategy for space, and we will be, I will need the ESA by my side." ESA director Aschbacher reciprocated, saying "I would really like to make the ESA the main agency, the go-to agency of the European Commission for all its flagship programmes." The ESA and EUSPA are now seen to have distinct roles and competencies, which will be officialised in the Financial Framework Partnership Agreement (FFPA). The ESA's focus will be on the technical elements of the EU space programmes, the EUSPA will handle the operational elements of those programmes.

Employment

As of 2023, Many other facilities are operated by national space agencies in close collaboration with the ESA. The ESA employs around 2,547 people, and thousands of contractors. Initially, new employees are contracted for an expandable four-year term, which is until the organisation's retirement age of 63. According to the ESA's documents, the staff can receive

myriad of perks, such as financial childcare support, retirement plans, and financial help when migrating. The ESA also prevents employees from disclosing any private documents or correspondences to outside parties. Ars Technica's 2023 report, which contained testimonies of 18 people, suggested that there is a widespread harassment between management and its employees, especially with its contractors. Since the ESA is an international organisation, unaffiliated with any single nation, any form of legal action is difficult to raise against the organisation.

Security incidents

On 3 August 1984, the ESA's Paris headquarters were severely damaged and six people were hurt when a bomb exploded. It was planted by the far-left armed Action Directe group.

On 14 December 2015, hackers from Anonymous breached the ESA's subdomains and leaked thousands of login credentials.

In late December 2025, ESA servers suffered a data breach, the extent and severity of which was disputed.

See also

[European integration § Space](#)

[European Space Security and Education Centre](#)

[Eurosace](#)

[List of European Space Agency programmes and missions](#)

[List of government space agencies](#)

[SEDS](#)

[Space Night](#)

[European Union matters](#)

Agencies of the European Union

Directorate-General for Defence Industry and Space

Enhanced co-operation

European Union Agency for the Space Programme

Notes

References

Further reading

External links

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